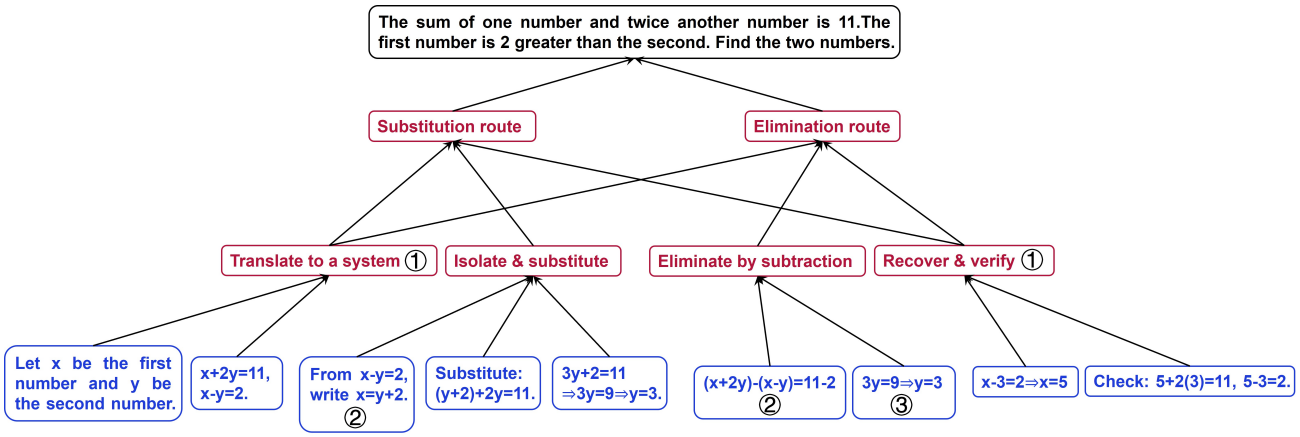

Figure H6. **A concrete algebra example illustrating skills, routing mechanisms, and identification conditions.** The **black box** denotes the problem descriptor $\mathbf{P}$. The **red boxes** denote latent selection variables. The **blue boxes** denote observed trace tokens. **Condition ①-Neighborhood coverage**: Shared modules, such as *"Translate to a system"* and *"Recover & verify"*, are accessible for both the substitution and elimination routes. **Condition ②-Local distinguishability**: Route-specific local traces, such as *"From x-y=2, write x=y+2"* and *"(x+2y)-(x-y)=11-2"*, help identify the route in use. **Condition ③-Restricted choice sets**: At each intermediate step, only a limited set of plausible next modules or trace continuations are available algebraically.

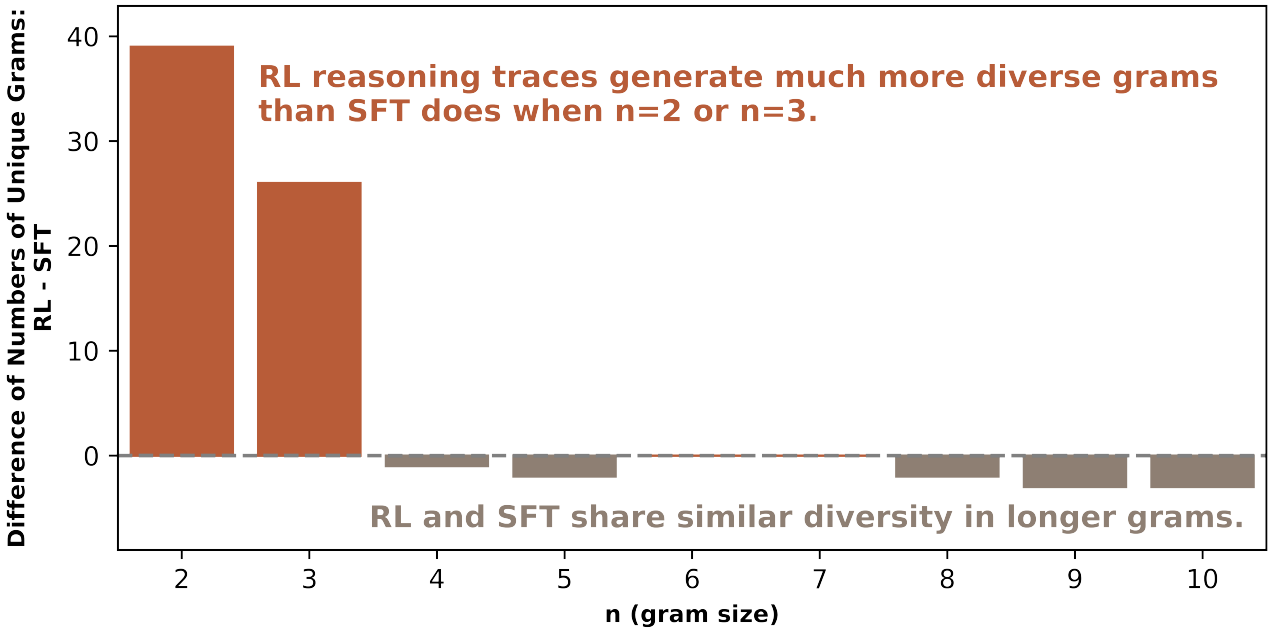

Figure H7. **Difference of numbers of unique grams between the RL and SFT model.** For the n-grams obtained in Appendix D (lines 979-983), we calculate the number of unique grams for both RL and SFT models, then visualize their differences here. The results show that (1) RL reasoning traces generate much more diverse grams than SFT does when $n = 2$ and $n = 3$; (2) RL and SFT reasoning traces share similar diversity in longer grams.